MIL-A-26891C(USAF)
7 July 1970
SUPERSEDING
MIL-A-26891B(USAF)
14 December 1961

MILITARY SPECIFICATION

AIR ELIMINATOR ASSEMBLY, RECESSED, AIRCRAFT FUEL FILTER

1. SCOPE

1.1 This specification covers an aircraft fuel filter recessed air eliminator assembly.

2. APPLICABLE DOCUMENTS

* 2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-V-3	Valves, Fittings, and Flanges (Except for Systems Indicated Herein), Packaging of

STANDARDS

Military

MIL-STD-100	Engineering Drawing Practices
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-808	Finishes, Protective, and Codes, for Finishing Schemes for Ground and Ground Support Equipment
MIL-STD-810	Environmental Test Methods
MIL-STD-831	Test Reports, Preparation of

Air Force-Navy Aeronautical

AN6230	Gasket, O-Ring Hydraulic

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 4930

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:30Z
Check the source to verify that this is the current version before use.

* 2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Bureau of Standards

Handbook H28 Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington D.C. 20402.)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 Components. The air eliminator assembly shall consist of the following major components:

Description	See Requirement
Air eliminator	3.7.1
Vapor trap	3.7.2
Support	3.7.3

* 3.3 Selection of standards and specifications. Standards and specifications for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 Materials. Materials shall be as specified herein. Materials not definitely specified shall be of the corrosion-resistant type or treated to resist corrosion due to fuels, salt spray, or atmospheric conditions likely to be met in storage or normal service and entirely suitable for the purpose. Materials shall be free from all defects and imperfections that might affect the serviceability of the finished air eliminator assembly.

3.5 Design and construction. The air eliminator assembly shall consist of an air eliminator designed for recessed mounting, a vapor trap, and a support designed and constructed to meet the performance requirements of this specification. The air eliminator assembly shall withstand the stresses, impacts, vibrations, and other conditions incident to shipping, storage, installation, and service.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:30Z
Check the source to verify that this is the current version before use.

3.5.1 Reliability. The air eliminator assembly shall be designed and constructed as specified herein and to successfully complete the tests of 4.4.3 with no failures preventing satisfactory performance.

3.5.2 Maintainability. The air eliminator assembly shall be designed and constructed as specified herein and to provide the following:

a. Minimum number of parts consistent with the reliability and performance specified herein.

b. Minimum amount of training and time necessary for assembly, disassembly, location of trouble sources, and maintenance including servicing. Where practical, parts and components shall be located or positioned for rapid and simple inspection and recognition of excessive wear or potential failure.

c. Permit adjustments, servicing, replacement of parts and components, and other maintenance with minimum disturbance to other equipment parts or components. Parts and components shall be located for ample and rapid access unless performance or reliability will be appreciably degraded by the accessible location. If engineering reasoning or data determines that in performing maintenance, physical or visual interference between items cannot be avoided, where practical, the item predicted to require the most maintenance shall be located for best accessibility.

d. Permit maintenance with general-purpose tools and equipment normally available commercially. Use of special-purpose tools and equipment shall be subject to procuring activity approval.

e. Minimum number of tools required for maintenance by design practices such as reducing the variety of bolthead sizes to the practicable minimum.

3.5.3 Foolproofness. Where improper installation of an item could cause malfunctioning of that item or the system in which it is installed, an unsymmetrical mounting means shall be provided. This mounting shall be so designed that the item can only be installed in its proper operating position.

3.6 Performance

3.6.1 Environmental conditions. The air eliminator assembly shall be capable of operating under and withstanding the following environmental conditions:

3.6.1.1 Temperature ranges

3.6.1.1.1 Operating temperatures. The air eliminator assembly shall be capable of performing as specified herein in ambient temperatures ranging from -65° to +130°F.

3.6.1.1.2 Storage temperatures. The air eliminator assembly shall not be damaged by storage in ambient temperatures ranging from -80°to +160°F for not less than 50 hours.

Source: https://assist.dla.mil -- Downloaded: 202640-20T10:30Z
Check the source to verify that this is the currentversion before use.

3.6.1.2 Humidity. The air eliminator assembly shall not be damaged by operation or storage in any relative humidity up to and including 100 percent, including conditions wherein condensation takes place in the form of water or frost.

3.6.1.3 Fungus. The air eliminator assembly shall not be damaged by exposure to moist fungus growth such as encountered in tropical and subtropical climates.

3.6.1.4 Sand and dust. The air eliminator assembly shall not be damaged by operation or storage in an atmosphere containing airborne sand and dust particles such as encountered in normal and desert operations.

3.6.1.5 Salt-fog. The air eliminator assembly shall not be damaged by exposure to salt-sea atmosphere.

3.6.2 Service pressure rating. The air eliminator, vapor trap, and associated piping and fittings shall be capable of withstanding a maximum working pressure of 150 psig and a hydrostatic test pressure of 225 psig.

3.6.3 Pressure rise. When subjected to a 600-gpm fluid flow, the pressure rise at the top of the vessel during the air discharge from the air eliminator mechanism, shall not exceed 5 psi. Liquid fuel discharge will not be permitted.

3.6.4 Supporting capability. With a 3/4-inch globe valve attached, the air eliminator assembly shall be capable of withstanding a 50-pound force applied in the horizontal plane and a 50-pound force applied in the vertical plane.

3.7 Details of components

3.7.1 Air eliminator. The eliminator shall consist of a housing designed for recessed mounting on a flange (as shown on figure 1) provided on a filter-separator and shall contain an internal float and fluid sealing mechanism.

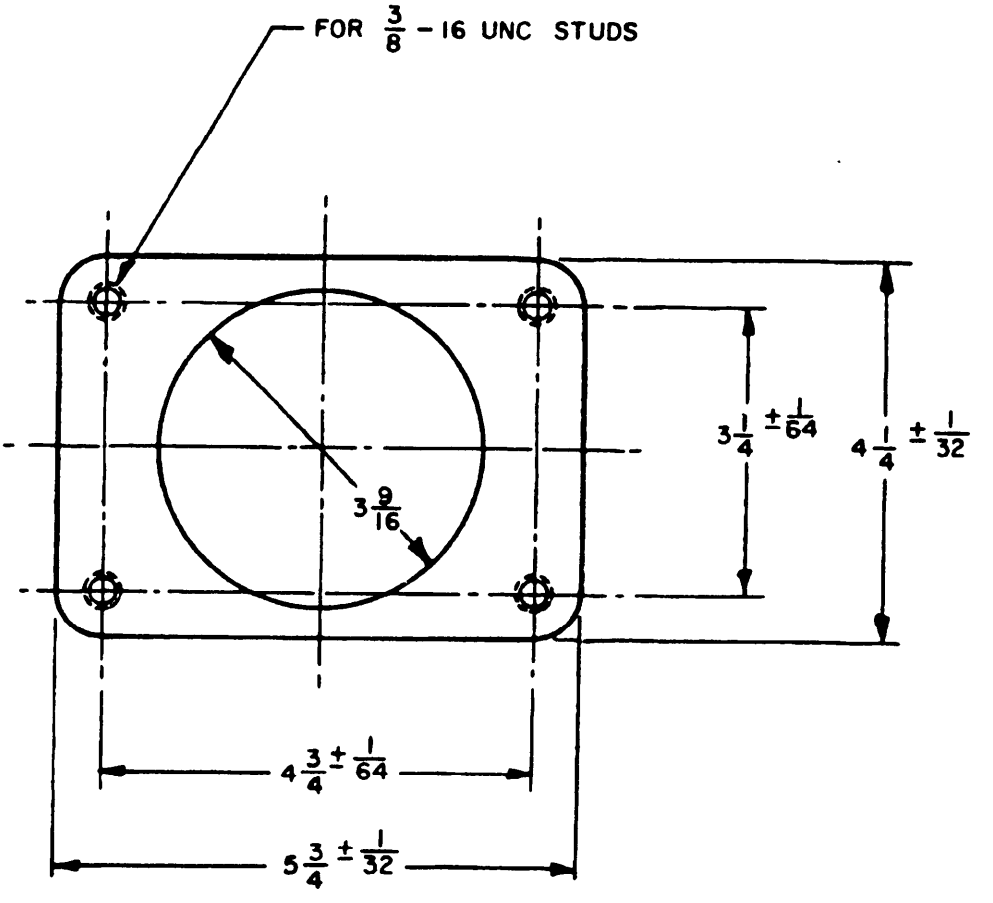
3.7.1.1 Configuration. The configuration of the air eliminator shall be as shown on figure 2. If an alternate design is used, only the outlet side of the air eliminator may differ from figure 2.

3.7.1.2 O-Ring. An O-ring shall be provided.

*3.7.1.3 Outlet. The air eliminator shall have one threaded outlet connection in a horizontal plane. Threads shall be 3/4-14 NPT internal.

3.7.1.4 Air removal. The air eliminator shall be so designed that while under a fuel pressure of 0.25 to 150 psig, it will discharge introduced air without discharge of liquid fuel.

Source: https://assist.dla.mil -- Downloaded: 202610-20T10:30Z
Check the source to verify that this is the currentversion before use.



DIMENSIONS IN INCHES

FIGURE 1. Filter Separator Flange

Source: https://assist.dla.mil -- Downloaded: 202610-20T10:30Z
Check the source to verify that this is the currentversion before use.

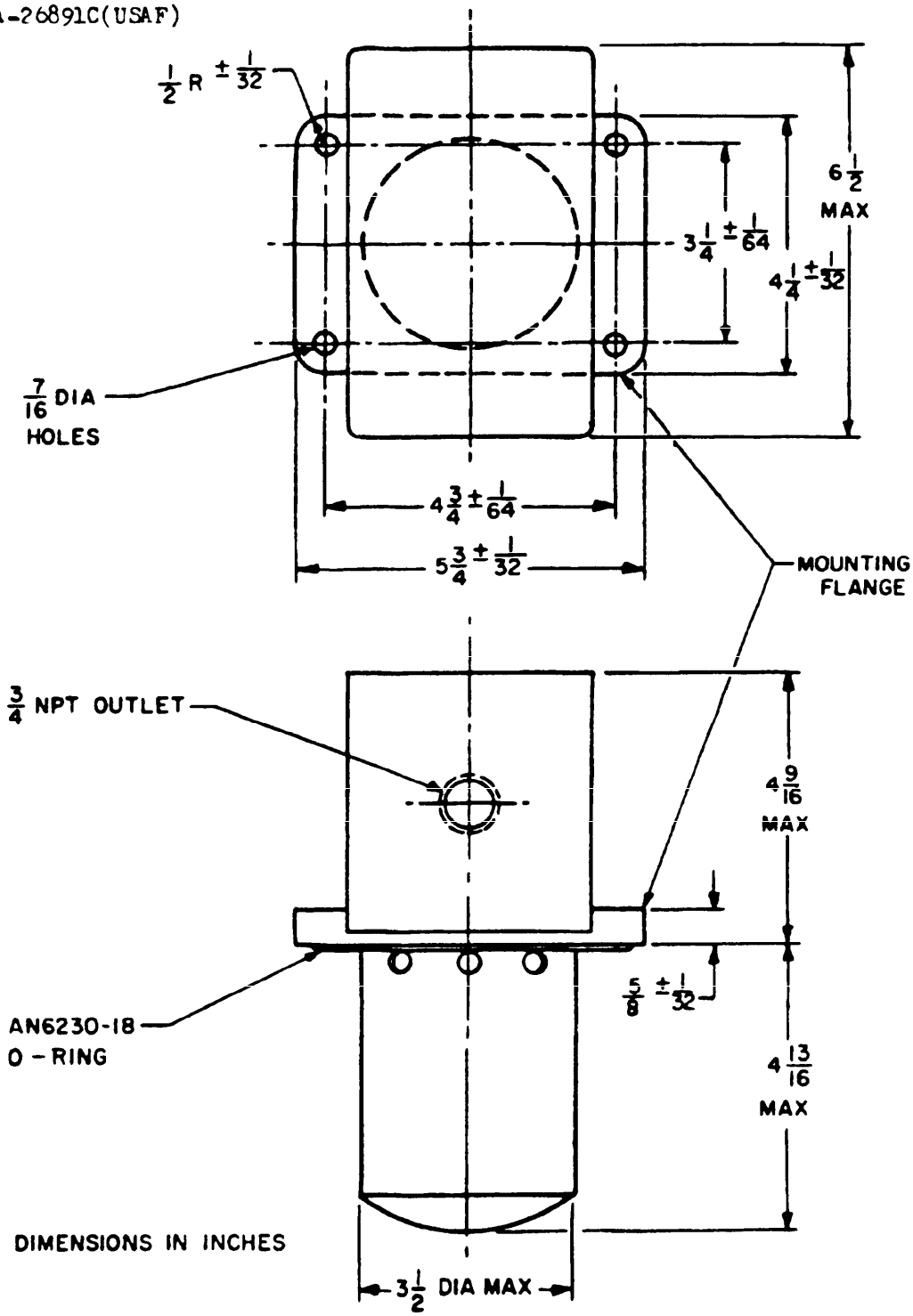


FIGURE 2. Recessed Air Eliminator

Source: https://assist.dla.mil -- Downloaded: 202610-20T10:30Z
Check the source to verify that this is the current version before use.

3.7.1.5 Valve pressure. The air eliminator shall be so designed that when completely filled with fuel, it will not leak when subjected to a pressure of 0.25 to 150 psig.

3.7.2 Vapor trap

3.7.2.1 Preferred design. The vapor trap shall prevent any liquid fuel or fuel in the form of a mist having a flow rate of 81 cfm which is emitted by the air eliminator, from being discharged to the atmosphere. The liquid fuel and the liquid fuel formed from the mist shall be returned to the filter-separator through the vapor trap inlet connection during the draining or defueling operation.

*3.7.2.1.1 Connections. The vapor trap inlet and outlet connections shall be 3/4-14 NPT internal.

3.7.2.1.2 Additional fittings. Additional fittings shall be provided to permit the vapor trap to be connected to the air eliminator without rotating either item.

*3.7.2.1.3 Sump capacity. The sump capacity of the vapor trap shall be not less than 20 cubic inches.

*3.7.2.2 Alternate design. An alternate vapor trap design in which the trap is incorporated as an integral part of the air eliminator will be acceptable. This design shall have a 3/4-14 NPT external outlet connection. The vapor trap shall pass all the requirements and tests specified herein.

3.7.3 Support. A support shall be provided for the vapor trap. The support shall be designed for attachment to the top of the air eliminator mounting flange and shall provide the necessary rigidity and strength for the vapor trap and 3/4-inch globe valve attached to the outlet connection. The support shall not extend below the horizontal plane of the air eliminator flange.

*3.8 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto.

*3.9 Screw and pipe threads. All screw and pipe threads shall be in accordance with Handbook H28.

3.10 Weight. The weight of the air eliminator assembly shall not exceed 16 pounds.

3.11 Finish. The air eliminator assembly shall be finished in accordance with MIL-STD-808, type I, color 13538.

Source: https://assist.dla.mil -- Downloaded: 202610-20T10:30Z
Check the source to verify that this is the current version before use.

3.12 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.13 Workmanship. Workmanship shall be of the highest grade throughout and shall be in accordance with good commercial practice for this type of equipment.

4. QUALITY ASSURANCE PROVISIONS

*4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the air eliminator assembly shall be classified as follows:

a. Preproduction testing . . . . . See 4.4

b. Acceptance tests . . . . . See 4.5

4.3 Test conditions. Insofar as is practicable, apparatus used in conjunction with the testing specified herein shall be of laboratory precision type and shall be calibrated at intervals properly spaced to insure laboratory accuracy.

4.4 Preproduction testing (see 6.2)

4.4.1 Test sample. One air eliminator assembly shall be subjected to the tests specified in 4.4.3.

*4.4.2 Test report. Upon completion of the preproduction tests, a preproduction test report in accordance with MIL-STD-831 shall be prepared and three copies of the report furnished to the procuring activity.

4.4.2.1 Reliability and maintainability information. The following information shall be submitted as an attachment accompanying the preproduction report (same quantity and quality of copies) or shall be included as part of that report:

a. All failures, maintenance, and other events recorded shall be identified by accumulated operating time, cycles, or position in the test procedure as appropriate. Test conditions during the failures or irregular operations identified shall be recorded.

Source: https://assist.dla.mil -- Downloaded: 202610-20T10:30Z
Check the source to verify that this is the current version before use.

b. Description of the engineering reasoning and of any tests conducted to determine assignable causes for all failures and irregular operations identified.

c. Description of the engineering reasoning behind any corrections made, to be made on production items, or proposed to be made and behind the predicted effectiveness of those corrections.

d. Test activity or contractor comments on item features or requirements that, if modified, should improve the item.

e. Test activity or contractor comments on field conditions or procedures to be avoided or cultivated to increase the reliability and useful life of the item.

4.4.3 Preproduction tests. The preproduction tests shall consist of all the tests described under 4.6.

4.5 Acceptance tests. The acceptance tests shall consist of the individual tests.

4.5.1 Individual tests. Each air eliminator assembly shall be subjected to the following tests as described under 4.6:

a. Examination of product . . . . . See 4.6.1

b. Service pressure rating test. . . . . See 4.6.2.

4.5.2 Sampling plan and tests. One air eliminator assembly of each one hundred units, or fraction thereof, produced shall be subjected to the following tests:

a. Air removal tests . . . . . See 4.6.3

b. Vapor trap test . . . . . See 4.6.4

c. Pressure rise test . . . . . See 4.6.5.

4.5.2.1 Rejection and retest. When one air eliminator assembly selected from a production run fails to meet the specification, air eliminator assemblies still on hand or produced later shall not be accepted until the extent and cause of failure have been determined and appropriately corrected. The contractor shall explain to the Government representative the cause of failure and the action taken to preclude recurrence. After correction, all of the tests shall be repeated.

4.5.2.2 Individual tests may continue. For production reasons, individual tests or other sampling plans may be continued pending the investigation of a sampling test failure. Final acceptance of the air eliminator assemblies on hand or produced later shall not be made until it is determined that all air eliminator assemblies meet all the requirements of the specification.

Source: https://assist.dla.mil -- Downloaded: 202610-20T10:30Z
Check the source to verify that this is the current version before use.

4.5.2.3 Defects in air eliminator assemblies already accepted. The investigation of a test failure could indicate that defects may exist in air eliminator assemblies already accepted. If so, the contractor shall fully advise the procuring activity of all the defects likely to be found and the method of correcting them.

4.6 Test methods

4.6.1 Examination of product. The air eliminator assembly shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, dimensions, and marking.

4.6.2 Service pressure rating test. The air eliminator, vapor trap, and associated piping and fittings shall be subjected to a hydrostatic pressure of 225 psig for not less than 30 minutes. There shall be no evidence of leakage nor impending failure as a result of this test.

4.6.3 Air removal tests

4.6.3.1 The air eliminator shall be mounted on a vessel, filled with fuel, and the pressure raised to 0.25 psig at an elevation equivalent to the top of the air eliminator. With this pressure maintained, air shall be introduced into the vessel inlet and the air-eliminator mechanism shall open and discharge air to atmosphere. This operation shall be repeated for 100 cycles. No liquid fuel discharge shall be permitted.

4.6.3.2 The air eliminator shall be mounted on a vessel, filled with fuel, and the pressure raised to 150 psig. With this pressure maintained, air shall be introduced into the vessel inlet and the air-eliminator mechanism shall open and discharge air to atmosphere. This operation shall be repeated for 100 cycles. No liquid fuel shall be permitted.

4.6.4 Vapor trap test. The vapor trap shall be subjected to a water mist with a velocity of 81 cfm. No liquid or mist discharge shall be permitted. This test shall be successfully conducted for 5 minutes or until 0.2 gallon of water is collected in the vapor trap. (Water is used for this test in lieu of fuel for safety purposes.)

4.6.5 Pressure rise test. The air eliminator shall be mounted on a drained vessel. Fluid flow at a rate of 600 gpm shall be diverted to the vessel. The pressure rise at the top of the vessel during the air discharge from the air-eliminator mechanism shall not exceed 5 psi. No liquid fuel discharge shall be permitted.

4.6.6 Valve pressure test

4.6.6.1 The air eliminator shall be mounted on a vessel, filled with fuel, and the pressure raised to 0.25 psig at an elevation equivalent to the top of the air eliminator. With this pressure maintained for a 30-minute period, no leakage shall occur.

Source: https://assist.dla.mil -- Downloaded: 202610-20T10:30Z
Check the source to verify that this is the current version before use.

4.6.6.2 The air eliminator shall be mounted on a vessel, completely filled with fuel, and the pressure raised to 150 psig. With this pressure maintained for a 30-minute period, no leakage shall occur.

4.6.7 Support test. The air eliminator assembly, with a 3/4-inch globe valve connected to the outlet connection, shall be installed in a rigid fixture in the operational position and subjected to the following tests:

4.6.7.1 The extreme end of the assembly (the globe valve) shall be subjected to a 50-pound force perpendicular to the assembly in the horizontal plane. There shall be no visible deflection as a result of this test.

4.6.7.2 The extreme end of the assembly (the globe valve) shall be subjected to a 50-pound downward force in the vertical plane. There shall be no visible deflection as a result of this test.

*4.6.8 Environmental conditions. The air eliminator assembly shall be subjected to the following environmental tests conducted in accordance with the specified procedures of MIL-STD-810. An engineering evaluation concerning the ability of the equipment to resist the exposure specified in 4.6.8.2 through 4.6.8.5 shall be considered as meeting the tests specified herein.

4.6.8.1 Temperature tests

*4.6.8.1.1 High temperature exposure test. The air eliminator assembly shall be subjected to high temperature in accordance with method 501, procedure I. The temperature of the chamber shall be reduced to 130°F and maintained until temperature stabilization of the test item is reached. The tests specified in 4.6.2, 4.6.3, and 4.6.5 shall be conducted.

*4.6.8.1.2 Low temperature exposure test. The air eliminator assembly shall be subjected to low temperature in accordance with method 502, procedure I. The assembly shall be cold-soaked at -80°F for 48 hours and then the temperature raised to -65°F and maintained until temperature stabilization of the test item is raised. The tests specified in 4.6.2, 4.6.3, and 4.6.5 shall be conducted.

*4.6.8.2 Humidity test. The air eliminator assembly shall be subjected to humidity in accordance with method 507, procedure I.

*4.6.8.3 Fungus test. The air eliminator shall be subjected to fungus in accordance with method 508, procedure I.

*4.6.8.4 Dust test. The air eliminator assembly shall be subjected to dust in accordance with method 510, procedure I.

*4.6.8.5 Salt-fog test. The air eliminator assembly shall be subjected to salt-fog in accordance with method 509, procedure I.

Source: https://assist.dla.mil -- Downloaded: 202640-20T10:30Z
Check the source to verify that this is the current version before use.

4.7 Inspection of the preservation, packaging, packing, and marking for shipment and storage. The inspection of the preservation, packaging, packing, and marking for shipment and storage shall be in accordance with section 5, or the documents specified therein.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. The air eliminator assembly shall be preserved, packaged, packed, and marked in accordance with MIL-V-3. (See 6.2)

6. NOTES

6.1 Intended use. The air eliminator assembly covered by this specification is intended for use as an integral part of a gasoline- or jet-fuel dispensing system, for installation on fuel servicing units.

*6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Location and conditions for preproduction testing (see 4.4)

c. Applicable levels of preservation and packaging, and packing required (see section 5).

*6.3 Abbreviations. In order to save space and to avoid the spelling out of repetitious words and phrases, the following abbreviations have been used in the text of this specification:

cfm	cubic feet per minute
gpm	gallons per minute
NPT	National taper pipe (thread)
psi	pounds per square inch
psig	pounds per square inch gage

*6.4 Identification of changes. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:30Z
Check the source to verify that this is the current version before use.

Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Air Force - 11

Reviewer:
Air Force - 82

Preparing activity:
Air Force - 11

Project No. 4930-F071

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:30Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:30Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD, FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

AFLC-WPAFB-OCT 67 2M

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:30Z
Check the source to verify that this is the current version before use.

ASD(ASNPS)
Wright-Patterson AFB, O 45433

UNITED STATES AIR FORCE
OFFICIAL BUSINESS

POSTAGE AND FEES PAID

ASD(ASNPS)
Wright-Patterson AFB, O 45433

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:30Z
Check the source to verify that this is the current version before use.